Exercise 4: SQL Practice Exercise: Joins, UNION, Filtering, Aggregates

Table 1: Employees - INNER JOIN

```
SELECT EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget,
       status
FROM Employees
INNER JOIN Projects
ON Employee.EmployeeID = Projects.EmployeeID;
```

EmployeeID	FirstName	LastName	Department
1	John	Doe	IT
1	John	Doe	IT
2	Alice	Smith	HR
3	Bob	Johnson	Finance
5	Emma	Wilson	Sales
6	Michael	Clark	Finance

LEFT JOIN

```
SELECT Employee.EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName
       Budget,
       Status
FROM Employees
LEFT JOIN Projects
ON Employees.EmployeeID = Projects.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL

RIGHT JOIN

```
SELECT ProjectID,
       ProjectName,
       Budget,
       Status,
       Employees.EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
FROM Employees
RIGHT JOIN
ON Employees.EmployeeID = Projects.EmployeeID;
```

ProjectID	ProjectName	Budget	Status
101	AI Development	100000	Completed
102	Employee Training	50000	Ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
105	Market Expansion	65000	Completed
106	Risk Management	80000	Pending

FULL OUTER JOIN

```
SELECT Employees.EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID
       ProjectName
       Budget
       Status
FROM Employees
FULL OUTER JOIN Projects.
ON Employees.EmployeeID = Projects.EmployeeID;
```

EmployeeID	First Name	Last Name	De